Rewrite

Rewriting reshapes the AIG before mapping without changing function

The idea

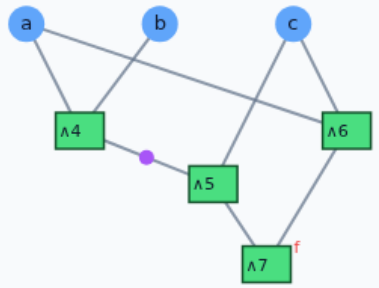
- Count nodes with complemented edges or AND children; flag AND(~x,~y) as NOR-friendly

Opportunities

REWRITE

Opportunities

Step 1 / 5 · opp · module02-03-rewrite-local



Explanation

Several AND nodes qualify as rewrite candidates.

- Complements
- AND kids

n=2

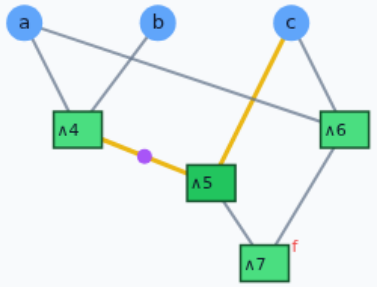
Opportunities

Both comp

REWRITE

Both comp

Step 2 / 5 · both · module02-03-rewrite-local



Explanation

AND(~, ~) is a NOR mapping hint.

- De Morgan

```
NOR2 later
```

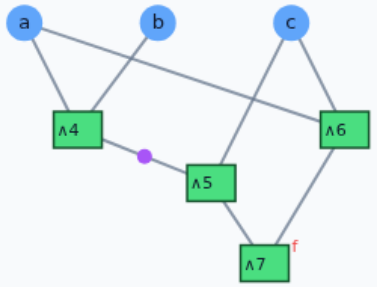
Both comp

Push

REWRITE

Push

Step 3 / 5 · push · module02-03-rewrite-local



Explanation

Inverter-push report lists those nodes.

- Lite rewrite

```
applied=0
```

Push

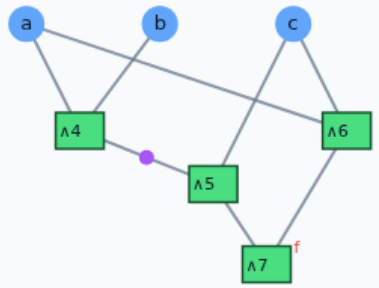
Preserve

REWRITE

Preserve

Step 4 / 5 · fn · module02-03-rewrite-local

f101=1



Explanation

Function f stays the same under rewrite bookkeeping.

- Equivalence

Toy

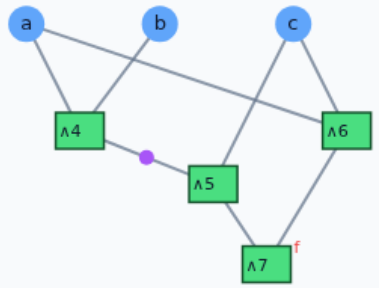
Preserve

Map later

REWRITE

Map later

Step 5 / 5 · map · module02-03-rewrite-local



Explanation

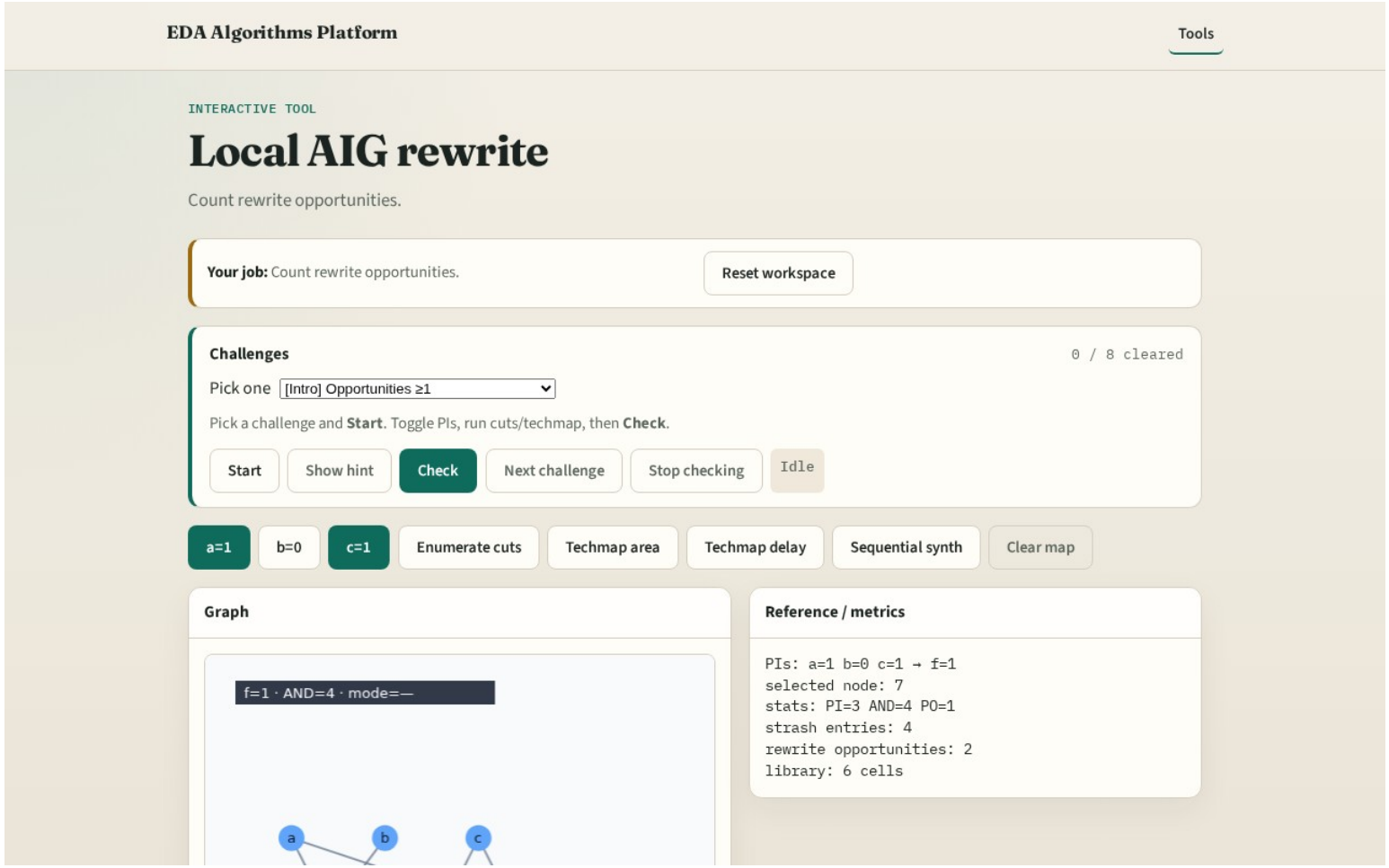
Techmap will use NOR2 when both comps appear.

- Library

```
Techmap
```

Map later

Browser lab track



Browser lab starter

Implement track

- Call local_rewrite_count and apply_inverter_push

Pitfalls

- Mutating function accidentally
- Counting PIs as rewrite targets

Your turn

- Next: k-feasible cuts